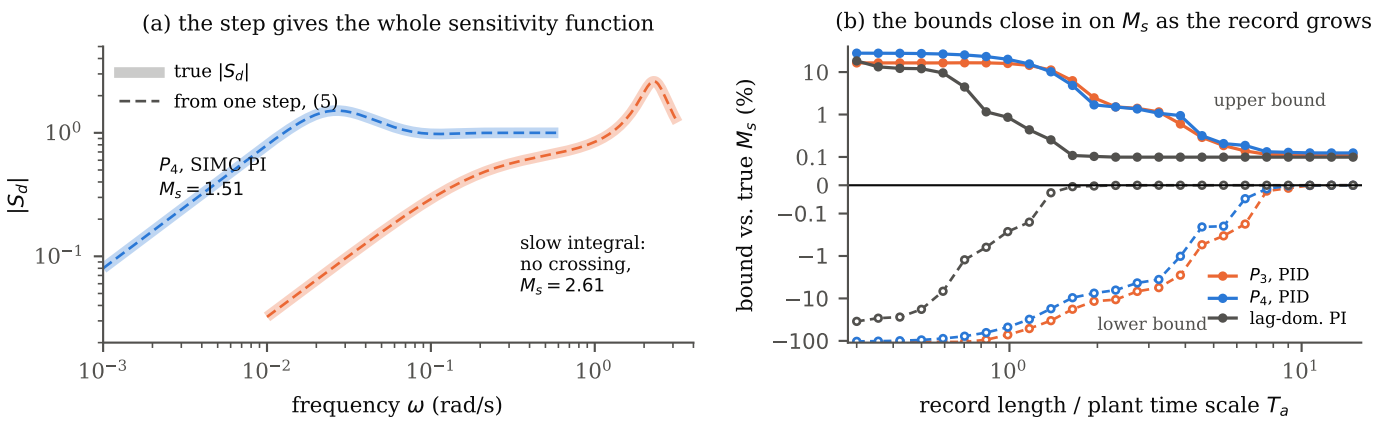
(a) the step gives the whole sensitivity function
true $|S_d|$
from one step, (5)
P_4, SIMC PI
$M_s = 1.51$
slow integral:
no crossing,
$M_s = 2.61$
$|S_d|$
10^0
10^{-1}
10^{-3}
10^{-2}
10^{-1}
10^0
frequency ω (rad/s)
(b) the bounds close in on M_s as the record grows
upper bound
lower bound
P_3, PID
P_4, PID
lag-dom. PI
bound vs. true M_s (%)
10
1
0.1
0
−0.1
−1
−10
−100
10^0
10^1
record length / plant time scale T_a